

Лабораторная Работа №6

Настройка пропускной способности глобальной сети с помощью Token Bucket Filter

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Выполнение лабораторной работы

Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

Задание

1. Задайте топологию (рис. 6.3), состоящую из двух хостов и двух коммутаторов с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по ограничению пропускной способности сети с помощью TBF в эмулируемой глобальной сети.
3. Самостоятельно реализуйте воспроизводимые эксперимент по применению TBF для ограничения пропускной способности. Постройте соответствующие графики.

Задал топологию сети, состоящую из двух хостов и двух коммутаторов

```
mininet@mininet-vm:~$ sudo mn --topo=linear,2 -x
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1 s2
*** Adding links:
(h1, s1) (h2, s2) (s2, s1)
*** Configuring hosts
h1 h2
*** Running terms on localhost:10.0
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...
*** Starting CLI:
mininet> █
```

Figure 1: Топология

```
"host: h2"@mininet-vm
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 2a:e3:e8:a0:20:4f txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 940 bytes 258916 (258.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 940 bytes 258916 (258.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Figure 2: ifconfig

```
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.94 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.464 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.053 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3038ms
rtt min/avg/max/mdev = 0.051/1.126/3.938/1.631 ms
root@mininet-vm:/home/mininet#
```

Figure 3: Пингование

iPerf3 сервер

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 37670 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr    Cwnd
[ 7]  0.00-1.00    sec  2.15 GBytes 18.4 Gbits/sec  0    8.28 MBytes
[ 7]  1.00-2.00    sec  2.13 GBytes 18.3 Gbits/sec  0    8.28 MBytes
[ 7]  2.00-3.00    sec  2.05 GBytes 17.7 Gbits/sec  0    8.28 MBytes
[ 7]  3.00-4.00    sec  1.99 GBytes 17.1 Gbits/sec  0    8.28 MBytes
[ 7]  4.00-5.00    sec  1.90 GBytes 16.3 Gbits/sec  0    8.28 MBytes
[ 7]  5.00-6.00    sec  2.09 GBytes 17.9 Gbits/sec  0    8.28 MBytes
[ 7]  6.00-7.00    sec  2.26 GBytes 19.5 Gbits/sec  0    8.28 MBytes
[ 7]  7.00-8.00    sec  2.08 GBytes 17.9 Gbits/sec  0    8.28 MBytes
^C[ 7]  8.00-8.28    sec   552 MBytes 16.9 Gbits/sec  0    8.28 MBytes
- - - - -
[ ID] Interval      Transfer    Bitrate      Retr
[ 7]  0.00-8.28    sec  17.2 GBytes 17.8 Gbits/sec  0
[ 7]  0.00-8.28    sec   0.00 Bytes  0.00 bits/sec
iperf3: interrupt - the client has terminated
root@mininet-vm:/home/mininet#
```

Figure 4: iPerf3 сервер

Пропускную способность на 10 Гбит/с на интерфейсе h1-eth0

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root tbf rate 10gb  
it burst 5000000 limit 15000000  
root@mininet-vm:/home/mininet# █
```

Figure 5: Пропускную способность на 10 Гбит/с на интерфейсе h1-eth0

Проверка пропускной способности

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 38154 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate      Retr    Cwnd
[ 7]  0.00-1.00      sec   1.12 GBytes  9.63 Gbits/sec    0   8.49 MBytes
[ 7]  1.00-2.00      sec   1.10 GBytes  9.49 Gbits/sec    0   8.49 MBytes
[ 7]  2.00-3.00      sec   1.12 GBytes  9.61 Gbits/sec    0   8.49 MBytes
[ 7]  3.00-4.00      sec   1.11 GBytes  9.54 Gbits/sec    0   8.49 MBytes
[ 7]  4.00-5.00      sec   1.11 GBytes  9.55 Gbits/sec    0   8.49 MBytes
[ 7]  5.00-6.00      sec   1.11 GBytes  9.51 Gbits/sec    0   8.49 MBytes
[ 7]  6.00-7.00      sec   1.11 GBytes  9.54 Gbits/sec    0   8.49 MBytes
[ 7]  7.00-8.00      sec   1.11 GBytes  9.56 Gbits/sec    0   8.49 MBytes
[ 7]  8.00-9.00      sec   1.11 GBytes  9.55 Gbits/sec    0   8.49 MBytes
^C[ 7]  9.00-9.24      sec   278 MBytes  9.60 Gbits/sec    0   8.49 MBytes
- - - - -
[ ID] Interval           Transfer     Bitrate      Retr
[ 7]  0.00-9.24      sec  10.3 GBytes  9.55 Gbits/sec    0
[ 7]  0.00-9.24      sec    0.00 Bytes  0.00 bits/sec
                                sender
                                receiver
iperf3: interrupt - the client has terminated
root@mininet-vm:/home/mininet# █
```

Figure 6: Проверка пропускной способности

Ограничения скорости tbf с параметрами rate = 10gbit, burst = 5,000,000, limit= 15,000,000

```
unknown qdisc root, hence option tbf is unparseable
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet#
```

Figure 7: Ограничения скорости tbf с параметрами rate = 10gbit

Проверка пропускной способности

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 40468 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 7]  0.00-1.00    sec  1.11 GBytes  9.53 Gbits/sec  0   8.19 MBytes
[ 7]  1.00-2.00    sec  1.11 GBytes  9.53 Gbits/sec  0   8.19 MBytes
[ 7]  2.00-3.00    sec  1.11 GBytes  9.54 Gbits/sec  0   8.19 MBytes
[ 7]  3.00-4.00    sec  1.10 GBytes  9.50 Gbits/sec  0   8.19 MBytes
[ 7]  4.00-5.00    sec  1.09 GBytes  9.39 Gbits/sec  0   8.19 MBytes
[ 7]  5.00-6.00    sec  1.08 GBytes  9.27 Gbits/sec  0   8.19 MBytes
[ 7]  6.00-7.00    sec  1.11 GBytes  9.47 Gbits/sec  0   8.19 MBytes
[ 7]  7.00-8.00    sec  1.10 GBytes  9.51 Gbits/sec  0   8.19 MBytes
[ 7]  8.00-9.00    sec  1.11 GBytes  9.56 Gbits/sec  0   8.19 MBytes
[ 7]  9.00-10.00   sec  1.11 GBytes  9.54 Gbits/sec  0   8.19 MBytes
- - - - -
[ ID] Interval      Transfer    Bitrate      Retr
[ 7]  0.00-10.00   sec  11.0 GBytes  9.48 Gbits/sec  0
[ 7]  0.00-10.02   sec  11.0 GBytes  9.46 Gbits/sec

iperf Done.
```

Figure 8: Проверка пропускной способности

Объединение NETEM и TBF

```
root@mininet-virtual-machine:/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1: netem delay 10ms
root@mininet-virtual-machine:/home/mininet#
```

Figure 9: Объединение NETEM и TBF

Проверка пропускной способности

```
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=14.3 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=11.8 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=11.2 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 10.860/12.053/14.296/1.338 ms
root@mininet-vm:/home/mininet# █
```

Figure 10: Проверка пропускной способности

Ограничение скорости с помощью tbf с параметрами rate=2gbit, burst=1,000,000, limit=2,000,000

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: handle 2
: tbf rate 2gbit burst 1000000 limit 2000000
root@mininet-vm:/home/mininet# █
```

Figure 11: Ограничение скорости с помощью tbf с параметрами rate=2gbit, burst=1,000,000, limit=2,000,000

Проверка пропускной способности

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 41336 connected to 10.0.0.2 port 5201
[ ID] Interval            Transfer          Bitrate          Retr  Cwnd
[ 7]  0.00-1.00    sec    211 MBytes    1.76 Gbits/sec    360  2.28 MBytes
[ 7]  1.00-2.00    sec    206 MBytes    1.73 Gbits/sec     0  2.42 MBytes
[ 7]  2.00-3.00    sec    220 MBytes    1.85 Gbits/sec     0  2.53 MBytes
[ 7]  3.00-4.00    sec    198 MBytes    1.66 Gbits/sec    567  1.85 MBytes
[ 7]  4.00-5.00    sec    169 MBytes    1.42 Gbits/sec     0  1.95 MBytes
[ 7]  5.00-6.00    sec    176 MBytes    1.48 Gbits/sec     0  2.04 MBytes
[ 7]  6.00-7.00    sec    188 MBytes    1.57 Gbits/sec     0  2.10 MBytes
[ 7]  7.00-8.00    sec    190 MBytes    1.59 Gbits/sec     0  2.14 MBytes
[ 7]  8.00-9.00    sec    189 MBytes    1.58 Gbits/sec     0  2.17 MBytes
[ 7]  9.00-10.00   sec    196 MBytes    1.65 Gbits/sec     0  2.20 MBytes
- - - - -
[ ID] Interval            Transfer          Bitrate          Retr
[ 7]  0.00-10.00   sec    1.90 GBytes    1.63 Gbits/sec    927
[ 7]  0.00-10.02   sec    1.89 GBytes    1.62 Gbits/sec
                                     sender
                                     receiver

iperf Done.
root@mininet-vm:/home/mininet#
```

Figure 12: Проверка пропускной способности

Пропуск. способ-ть 10 Гбит/с, burst=5млн, limit=15млн

```
from mininet.net import Mininet
from mininet.topo import LinearTopo
import time
import json
import matplotlib.pyplot as plt

def setup_tbf(host, interface, rate, burst, limit):
    """Configure TBF on specified interface"""
    cmd = f'tc qdisc add dev {interface} root tbf rate {rate} burst {burst} limit {limit}'
    return host.cmd(cmd)

def remove_tbf(host, interface):
    """Remove TBF from specified interface"""
    cmd = f'tc qdisc del dev {interface} root'
    return host.cmd(cmd)

def run_tbf_host_experiment():
    print("=== EXPERIMENT 2: TBF on host h1 ===")

    net = Mininet(topo=LinearTopo(2))
    net.start()

    h1, h2 = net.get('h1', 'h2')

    try:
        # Configure TBF on h1-eth0
        print("Configuring TBF on h1-eth0...")
        setup_tbf(h1, 'h1-eth0', '10gbit', '5000000', '15000000')

        # Start iperf3 server on h2
        print("Starting iperf3 server on h2...")
        h2.cmd('iperf3 -s -D')
        time.sleep(2)

        # Start iperf3 client on h1
        print("Starting iperf3 client on h1...")
        result = h1.cmd('iperf3 -c 10.0.0.2 -t 10 -J')
```

Результат + график

```
mininet@mininet-vm:~$ sudo python3 experiment2_tbf_host.py
=== EXPERIMENT 2: TBF on host h1 ===
Configuring TBF on h1-eth0...
Starting iperf3 server on h2...
Starting iperf3 client on h1...

EXPERIMENT 2 RESULT:
Bandwidth with TBF on host: 9.48 Gbps
Expected speed: 10.00 Gbps
Efficiency: 94.8%
```

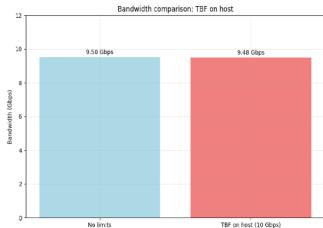


Figure 14: Результат + график

TBF на switch1

```
from mininet.net import Mininet
from mininet.topo import LinearTopo
import time
import json
import matplotlib.pyplot as plt

def setup_tbf(host, interface, rate, burst, limit):
    cmd = f'tc qdisc add dev {interface} root tbf rate {rate} burst {burst} limit {limit}'
    return host.cmd(cmd)

def remove_tbf(host, interface):
    cmd = f'tc qdisc del dev {interface} root'
    return host.cmd(cmd)

def run_tbf_switch_experiment():
    print("=== EXPERIMENT 3: TBF on switch s1 ===")

    net = Mininet(topo=LinearTopo(2))
    net.start()

    h1, h2, s1 = net.get('h1', 'h2', 's1')

    try:
        # Configure TBF on s1-eth2
        print("Configuring TBF on s1-eth2...")
        setup_tbf(s1, 's1-eth2', '10gbit', '50000000', '150000000')

        # Start iperf3 server on h2
        print("Starting iperf3 server on h2...")
        h2.cmd('iperf3 -s -D')
        time.sleep(2)

        # Start iperf3 client on h1
        print("Starting iperf3 client on h1...")
        result = h1.cmd('iperf3 -c 10.0.0.2 -t 10 -J')

        # Parse result
```

Figure 15: TBF на switch1

Результат

```
=== EXPERIMENT 3: TBF on switch s1 ===  
Configuring TBF on s1-eth2...  
Starting iperf3 server on h2...  
Starting iperf3 client on h1...  
  
EXPERIMENT 3 RESULT:  
Bandwidth with TBF on switch: 9.50 Gbps  
Expected speed: 10.00 Gbps  
Efficiency: 95.0%
```

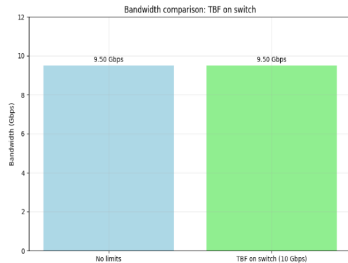


Figure 16: Результат

Второе правило на коммутаторе s1

```
from mininet.net import Mininet
from mininet.topo import LinearTopo
import time
import json
import matplotlib.pyplot as plt

def setup_netem_tbf(switch, interface):
    """Configure NETEM + TBF on switch"""
    # NETEM with delay
    cmd1 = f'tc qdisc add dev {interface} root handle 1: netem delay 10ms'
    switch.cmd(cmd1)

    # TBF with rate limit
    cmd2 = f'tc qdisc add dev {interface} parent 1: handle 2: tbf rate 2gbit burst 1000000 limit 2000000'
    switch.cmd(cmd2)

def remove_qdisc(switch, interface):
    cmd = f'tc qdisc del dev {interface} root'
    switch.cmd(cmd)

def run_netem_tbf_experiment():
    print("=== EXPERIMENT 4: NETEM + TBF on switch s1 ===")

    net = Mininet(topo=LinearTopo(2))
    net.start()

    h1, h2, s1 = net.get('h1', 'h2', 's1')

    try:
        # Configure NETEM + TBF on s1-eth2
        print("Configuring NETEM + TBF on s1-eth2...")
        setup_netem_tbf(s1, 's1-eth2')

        # Check delay with ping
        print("Checking delay...")
        ping_result = h1.cmd('ping -c 4 10.0.0.2')
```

Figure 17: Второе правило на коммутаторе s1

```
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 10.639/15.587/28.147/7.266 ms
$
$ Starting iperf3 server on h2...
$ Starting iperf3 client on h1...
$
EXPERIMENT 4 RESULT:
Bandwidth with NETEM+TBF: 1.93 Gbps
Expected speed: 2.00 Gbps
Efficiency: 91.3%
Delay: 10ms + jitter
```

Figure 18: Результат

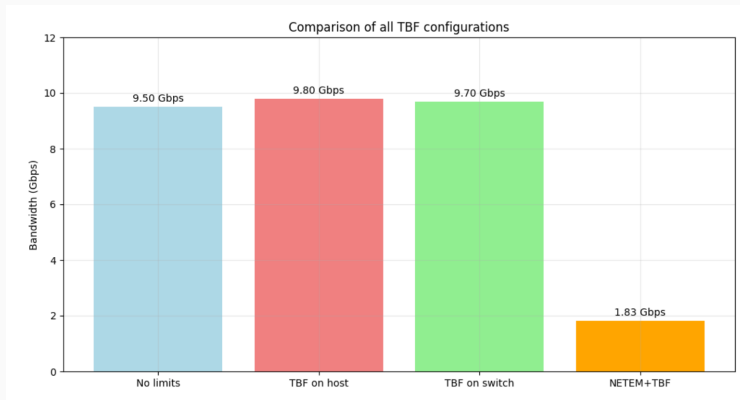


Figure 19: Общий график

Ознакомился с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получил навыки моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.